Answer all the questions

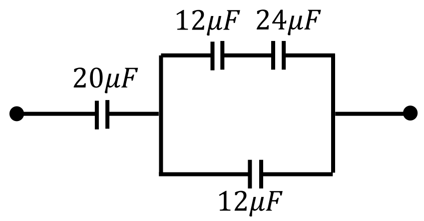
1. How much charge is on each plate of a 4.00-μF capacitor when it is connected to a 12.0-V battery?

[$48.0\mu C$]

2. Two conductors having net charges of +10.0 μC and –10.0 μC have a potential difference of 10.0 V between them. Determine the capacitance of the system.

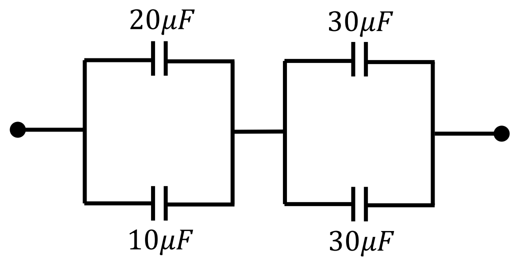
[$1.0\mu F$]

3. What is the equivalent capacitance of the combination shown?



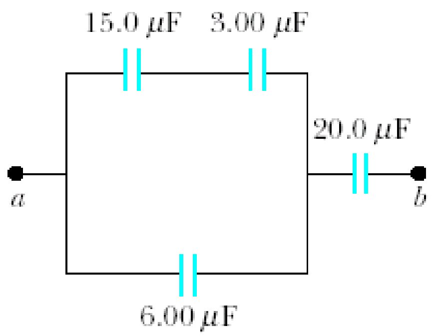
[$10\mu F$]

4. What is the equivalent capacitance of the combination shown?



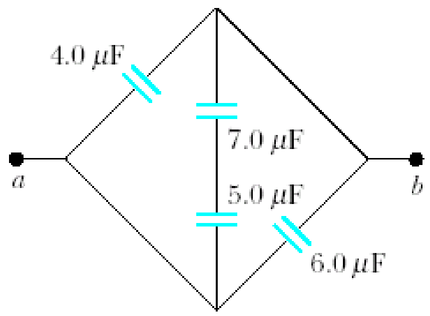
[$20\mu F$]

5. Four capacitors are connected as shown in figure below. Find the equivalent capacitance between points *a* and *b*.



[$5.96\mu F$]

6. Find the equivalent capacitance between points *a* and *b* in the combination of capacitors shown in Figure below.



[$12.9\mu F$]

7. A 20 μF capacitor has a charge of 240 μC is connected to a 50 kΩ resistor. Calculate the time constant.

[$1.0s$]

8. Sketch and label the graph $Q - t$ of a capacitor during
   I. charging
   II. discharging

9. A $12\mu F$ capacitor is charged to a potential difference of $6V$. It is then disconnected and discharges through $20k\Omega$ resistor. Calculate
   I. The time constant
   II. The charge remained in the capacitor after $0.5s$.

[$0.24s$, $8.97 \times 10^{-6}C$]

10. A $1000\mu F$ capacitor is connected in series with a $2.0k\Omega$ resistor to a battery of $4.0V$. Calculate the charge stored in the capacitor after $5.0s$.

[$3.67 \times 10^{-3}C$]